

Amr Elkholy, MS

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EDUCATION

New York University

M.S. in Quantitative Management

Relevant Coursework: Machine Learning, Statistical Modeling, Predictive Analytics, Data Visualization, Optimization & Decision Analysis, Database Systems, Financial Analytics

New York, NY

5/2026

Massachusetts Institute of Technology (MIT) - Applied AI & Data Science Program

2026

Focus: Machine Learning & Deep Learning & Recommendation systems (neural networks, predictive modeling in Python)

Rutgers Business School

B.A. in Business Administration, Data Analytics Concentration

New Brunswick, NJ

12/2021

WORK EXPERIENCE

New York University — Costing & Analysis

Senior Data Analyst

New York, NY

5/2022 – 5/2026

- Developed the data infrastructure for the university's federally negotiated Facilities & Administrative (F&A) overhead rate model, integrating financial, payroll, facilities, depreciation, and space management data into a centralized SQL Server database. Built ETL pipelines, SQL transformations, data validation processes, and Python automation to support accurate, auditable annual submissions that maintained the university's 65% negotiated F&A rate.
- Architected a modern data platform using Python, Docker, Apache Airflow, SQL, dbt, and Git, designing normalized relational databases, SQL data marts, and automated ELT pipelines to support enterprise analytics, self-service reporting, and external regulatory reporting.
- Designed and managed a centralized SQL reporting hub consisting of 200+ automated reports supporting executive leadership, federal agencies (NASA, NSF, NIH, DoD), and global university ranking submissions.
- Implemented automated data quality and reconciliation frameworks using SQL and Python, improving reporting accuracy and ensuring compliance with federal reporting requirements.
- Built financial forecasting models and executive reporting solutions using OBIEE and Tableau, enabling leadership to optimize research funding decisions and reduce sponsor receivables from \$18M to \$3M.
- Designed and delivered executive Power BI dashboards that translated complex research spending data into actionable insights for leadership and Principal Investigators, supporting a 30% increase in research expenditures over a two-year period.
- Developed a Power BI dashboard using advanced DAX to map research activity to individual rooms, enabling room-level cost-benefit analysis, space utilization optimization, and F&A recovery reporting. Contributed to a 30% improvement in research space utilization.
- Co-developed a Microsoft Access invoicing and reporting platform using VBA, SQL, and Microsoft Access, integrating with the university's Enterprise Data Warehouse (via OBIEE). Automated invoice generation and receivables reporting, reducing invoice processing time from approximately 1 hour to 5 minutes per invoice while supporting 20+ daily users.
- Developed enterprise reporting solutions supporting executive leadership, 1,000+ Principal Investigators, and research administrators across the university.
- Served as Lead Business Analyst, collaborating with IT and external consultants to deliver two grant management automation tools for closeout processing and compliance monitoring. Spearheaded the integration of pre-award and post-award systems, aligning budgets with expenditures to improve grant lifecycle management.
- Managed cross-functional analytics and technology initiatives using Jira and Asana, prioritizing competing workstreams, coordinating stakeholders, driving change management, and mentoring junior analysts.
- Established documentation, testing, and data governance standards by authoring 20+ process documents, defining business rules and KPI definitions, improving knowledge transfer, audit readiness, and operational consistency.

Precision Pain & Spine Institute

Financial / Billing Analyst

New Jersey

9/2020 – 4/2022

- Designed Power BI dashboards to trend revenue, provider productivity, and case profitability across a multi-specialty physician group (pain management, chiropractic, physical therapy), partnering with leadership on KPI definition and executive reporting.
- Managed revenue cycle analytics for the 10-physician group, optimizing billing workflows and reducing claim denial rates.
- Resolved \$17M in outstanding debit balances through systematic analysis, variance investigation, and targeted collection strategies.
- Led appeals for 2,000+ insurance arbitration cases, achieving a 50% profitability rate via data-driven prioritization and analytical modeling.
- Led a financial system implementation, reconciling \$20M+ across 15,000+ transactions while ensuring accuracy, regulatory compliance, and audit readiness.

Precision Pain & Spine Institute

Billing Specialist

New Jersey

12/2018 – 8/2020

- Captured \$10M+ in revenue through charge-capture and claim-submission workflow optimization, ensuring compliance with billing regulations.
- Reduced insurance claim backlog by 20% within one year via Excel-based tracking systems and targeted denial resolution.
- Improved 60-day collection rates by 20% across all insurance payers through a structured project-planning roadmap and process improvements.
- Enhanced revenue recognition accuracy by 25% through database optimization using advanced Excel (pivot tables, VLOOKUP).

SKILLS

- Data Engineering: SQL Server, Python, Docker, Airflow, dbt, Git, ETL/ELT.
- Business Intelligence: Power BI, Tableau, OBIEE, DAX, Excel
- Analytics: Forecasting, F&A modeling, financial analysis.
- Leadership: Business analysis, stakeholder management, mentoring.
- Governance: Data quality, documentation, compliance.